

Generation Interconnection System Impact Study Report

For Queue Project Q23-2323

Holly Springs 138 kV
May 27, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q23-2323. The Interconnection Customer proposes a 99.5 MW solar photovoltaic facility interconnecting at the Holly Springs 138 kV substation in the Pennsylvania area. The request was received on June 21, 2023. The study evaluated system intact and single contingency conditions using the regional planning case, including 671 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,880,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Apollo - Holl Springs 138.0 kV	St Hedwig - Snapper Creek 138.0 kV	113.4	248	76.2
Apollo - Seven Springs 138.0 kV	St Hedwig - Snapper Creek 138.0 kV	104.5	259	76.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Apollo - Holl Springs 138.0 kV	Rebuild line with higher-capacity conductor	\$3,775,000	50% (\$1,889,000)
Apollo - Seven Springs 138.0 kV	Reconductor line	\$1,981,000	50% (\$991,000)
	Total allocated to project		\$2,880,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.